

Bio 40C: Acid-Base Balance

Jeff Schinske



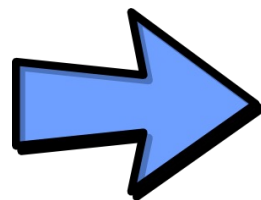


Bio 40C Proposed Norms

- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share with a partner...

1. Name
2. Something important to know about you as a person
3. What, if any, Halloween traditions do you have?



Person whose birthday is coming up next goes first



Debriefing Last Week

○ Last Week's Canvas Work

- "Learning about the mapping of the nephron, step by step, was helpful. I feel so much more confident now and enjoyed doing the self-explanation followed with a review quiz."
- "I really enjoyed the visualization tour, it was fun trying to explain what kind of environment I'd be in if I was a little molecule exploring each nephron."
- "I realized that I didn't know as much about the kidneys as I thought I did. There were still a lot of in-depth things to learn about such as exactly what gets re-absorbed and secreted at the various stages."

○ Application Question 7 (with the two graphs)

- Refer to last week's lecture recording for tips!



From our course calendar:

Date		Lecture Topics (Physiology)
Sept	26	Introduction to Bio 40C & Digestion
Oct	3	Digestion & Metabolism
	10	Urinary System
	17	Urinary System
	24	Fluid & Solute Balance
	31	Endocrine System <i>1st Deadline for Revising Application Questions</i>

- Application questions represent our core learning goals each week!
- Your application question responses can be revised any time until our final exam.
- However, I would like everyone to have achieved a “mostly complete” response on the questions so far!
 - Revise any 6/10 or 3/10 application questions by the end of October 31st to receive bonus, mid-quarter credit!
 - Some points will be lost if you have 6/10 or 3/10 scores on application questions after Oct. 31st



A Brief Preview of This Week

- Today
 - Finishing our urinary system case study
 - Considering the ways our bodies regulate pH
- Lab This Week
 - Group activities to review our last four weeks of lab
- Online Lecture Work This Week
 - Looking at sports drinks as a way of understanding how we balance water and salts in our body fluids

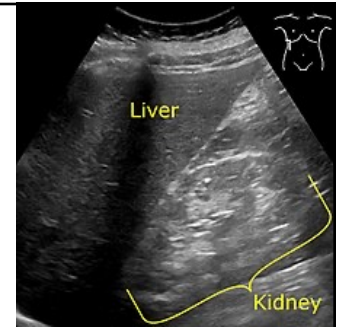


What will you be able to do today?

After class, you will be able to:

- Explain a medical case using an A&P perspective.
- Explain to a hypothetical friend the physiological and health impacts of drinking "alkaline water."

Case 1: Analysisia



Analysisia remarks, “Mostly I’m doing pretty well! I’ve been trying to take care of myself by getting in some exercise and watching my diet, just like you advised me last time. I have to admit, though, that I find myself needing to use the restroom quite a lot these days. Maybe I’m just getting to be an old lady. I also have occasional, mild headaches, but again, I don’t think I can complain too much.”

You look at Analysisia’s record and recall that, 5 years ago, she was diagnosed with diabetes and has been in treatment for that condition. That leads you to perform some tests. You also notice that Analysisia’s legs appear puffy.

You receive the following results for Analysisia:

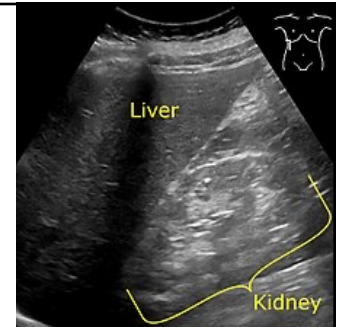
Test	Analysisia’s Results
Urinalysis	Albumin: 100 mg/24hr
Blood Creatinine	eGFR of 60 ml/min
Ultrasound	See picture
Blood Pressure	High

Step 2: Diagnosis

Perform additional research to interpret these results.

- What is being measured in the urinalysis and blood tests?
- What condition does Analysisia have?
- How do the urinalysis, blood, and ultrasound results relate to Analysisia’s condition?

Case 1: Analysisia



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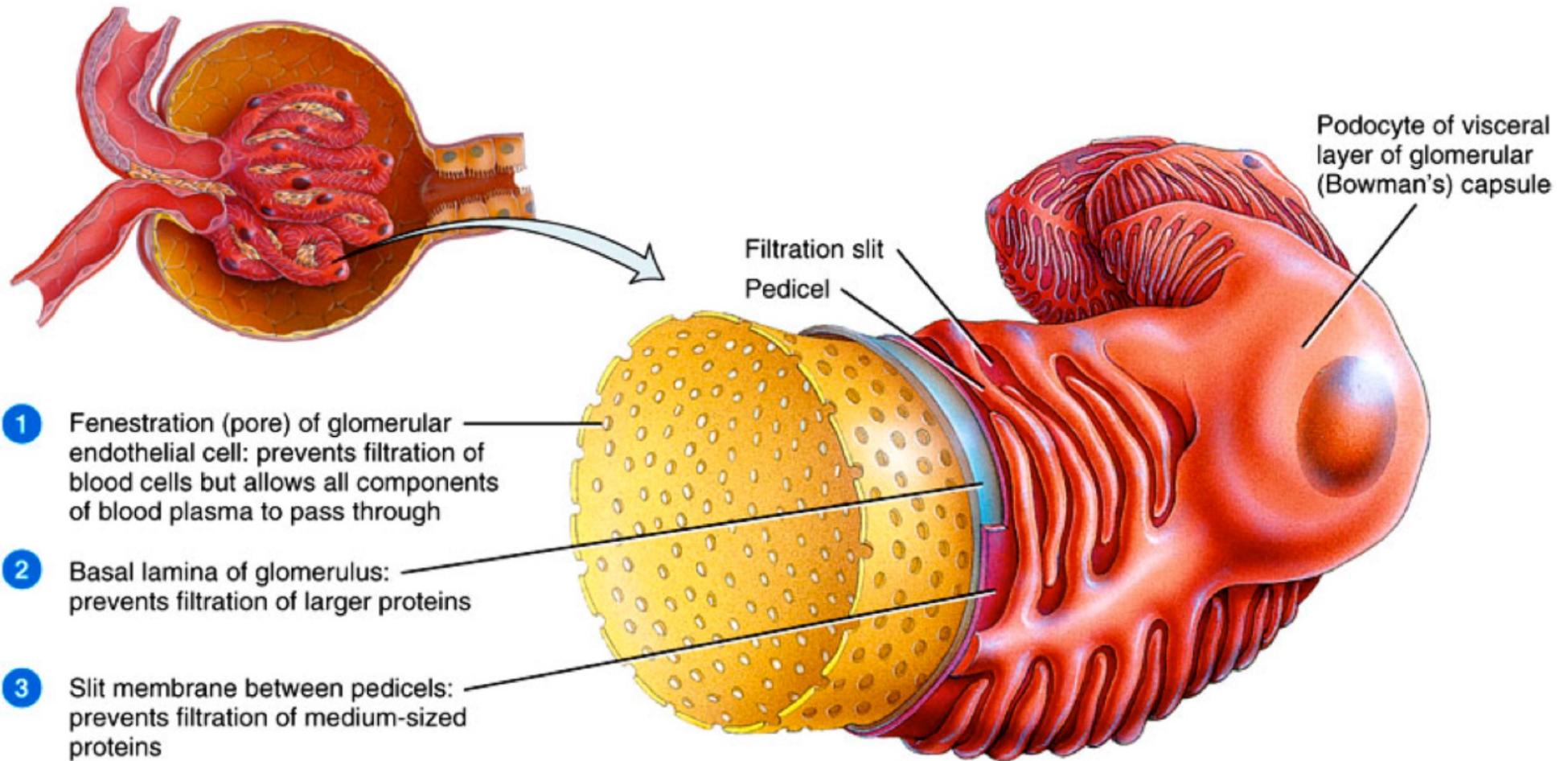
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Step 3: Cellular Explanation

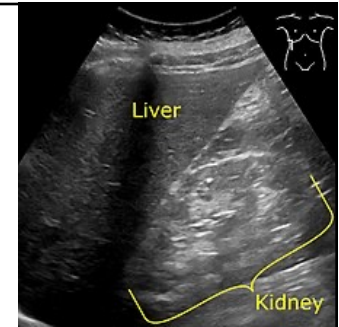
Perform additional research on this condition.

- What cells or other microscopic structures are most impacted?**
- Make a rough diagram of a glomerulus and the abnormalities Analysisia’s glomeruli would have.**

What gets through the filter?



Case 1: Analysisia



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Step 4: Treatment

Perform additional research on this condition.

- What will be the likely treatment for Analysisia?
- From a physiology perspective, how will that treatment work for her?



The Popularity of “Alkaline Water”

The New York Times

UPTICK

Alkaline Water Makes a Big Splash

By Alex Williams

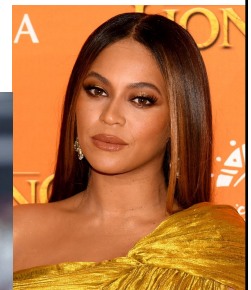
Nov. 16, 2018



The Hype: Tom Brady drinks it. Beyoncé is reportedly hooked. Cycling studios on both coasts can't keep enough of it on the shelves. In our never-ending quest for the ultimate form of hydration, [alkaline water](#) has emerged as the eau du jour, touted as “energizing,” “detoxifying” and a cure-all for ailments that seem to afflict people who congregate at yoga studios and juice bars.

Want to hydrate like Beyoncé? Pick up this alkaline water bottle that just went on sale.

Alkaline water does nothing: Tom Brady and Beyonce swear by it - but scientists say it's just a fad





Recent Popularity of “Alkaline Water”

Have you heard of alkaline water before today?

What do you know about it?

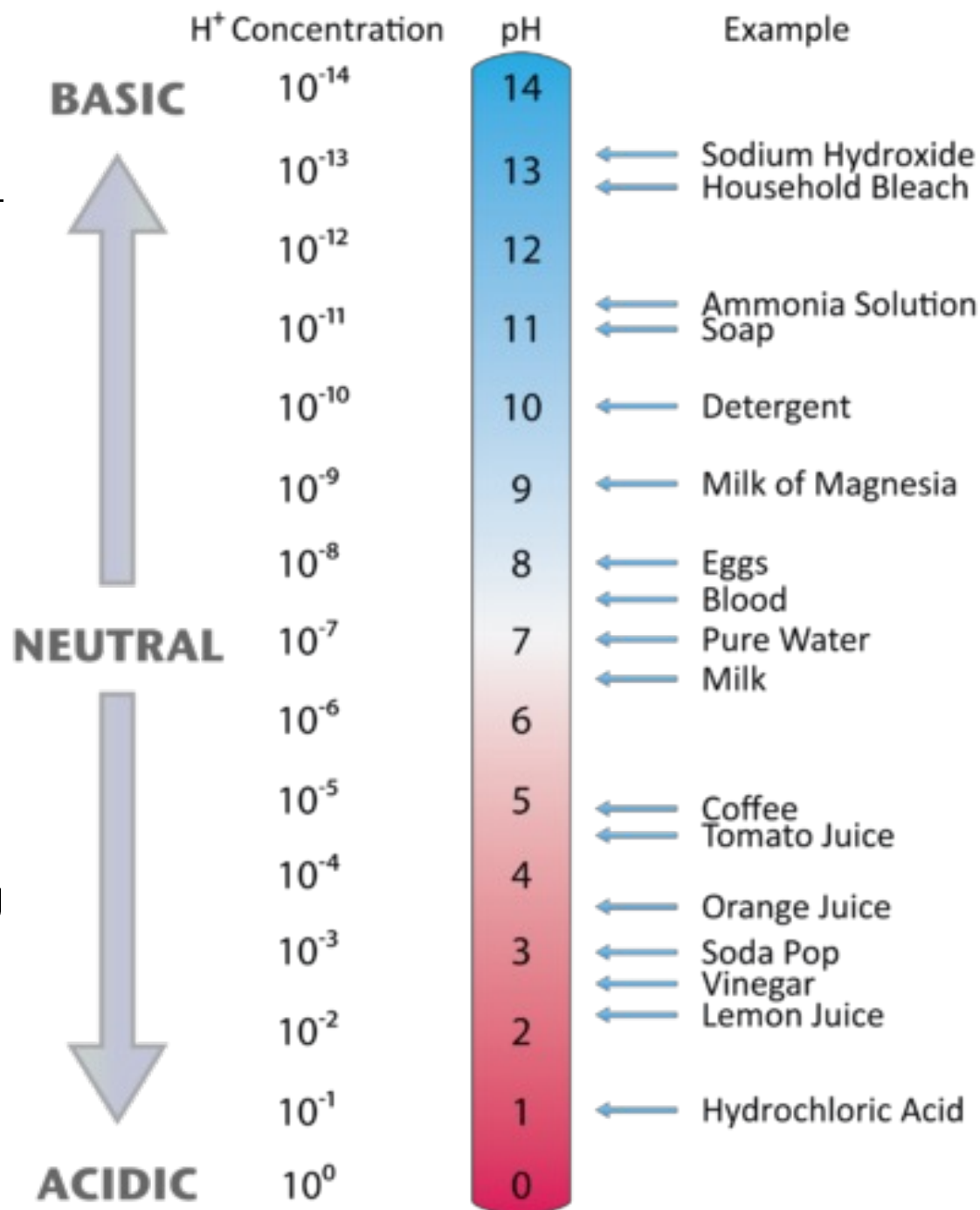
pH is a measurement scale (not a substance) that tells us the amount of H^+ in water

pH Scale

Base: Chemical capable of reducing H^+ and making water more basic

Buffer: Solution of water with chemicals that easily accept/release H^+ and keep pH stable

Acid: Chemical capable of releasing H^+ and making water more acidic



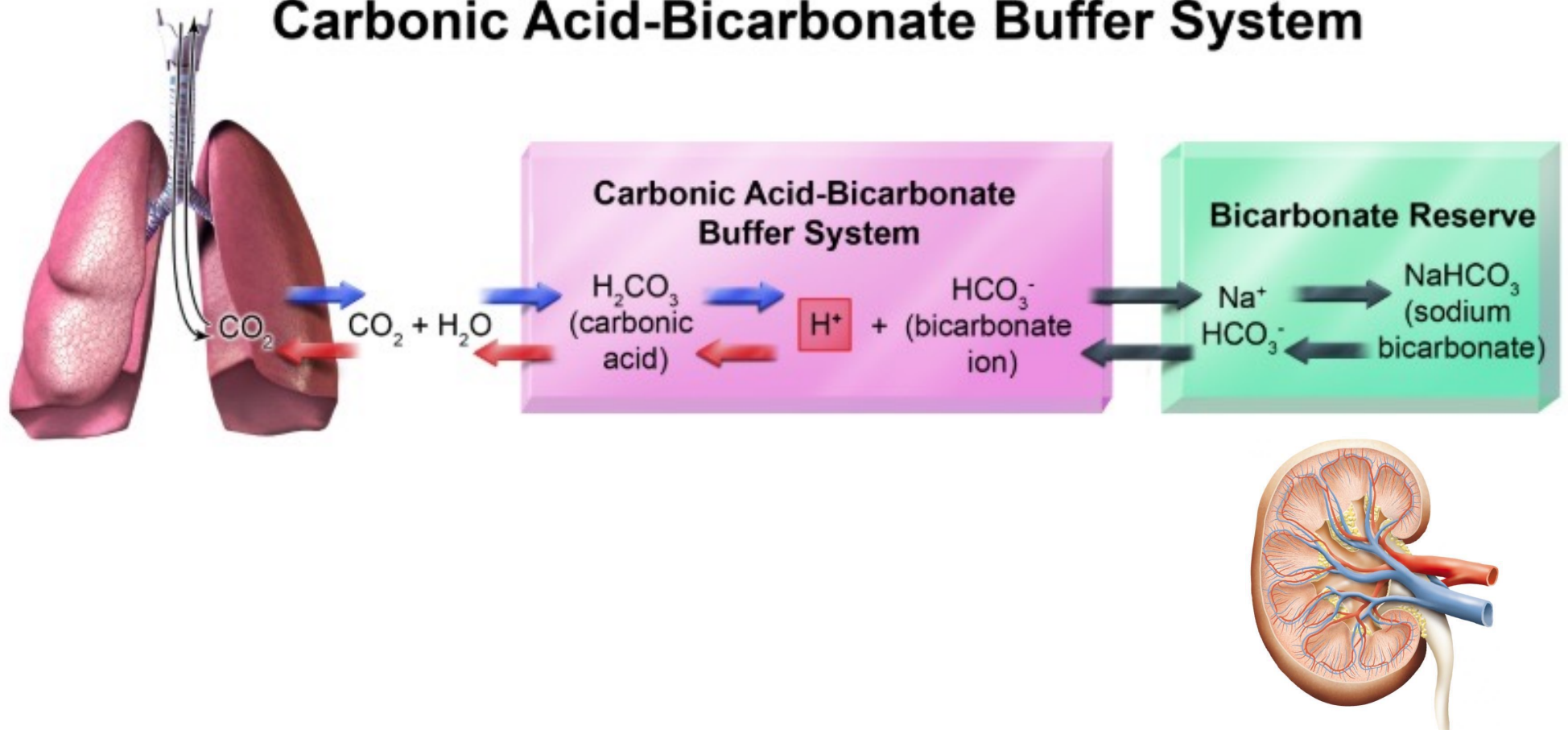


On the merits, or lack thereof, of alkaline water...

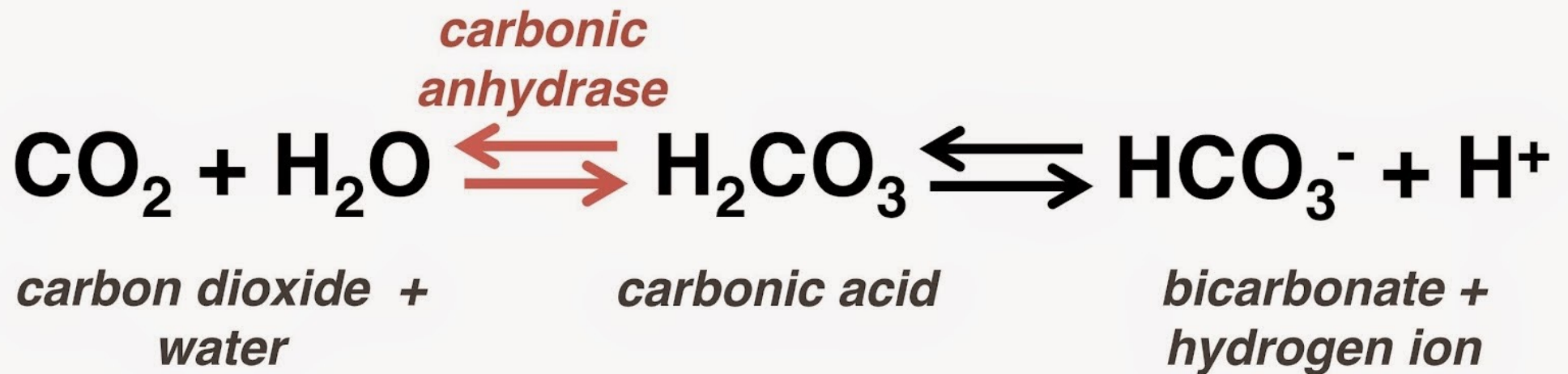
<https://www.youtube.com/watch?v=1CBE-5Ot4jU>

Keeping Bodily pH Stable

Carbonic Acid-Bicarbonate Buffer System

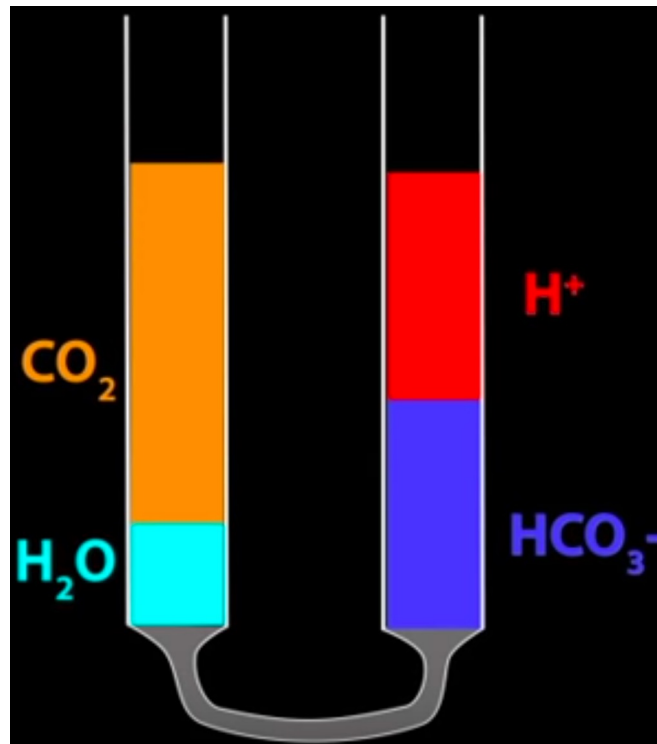
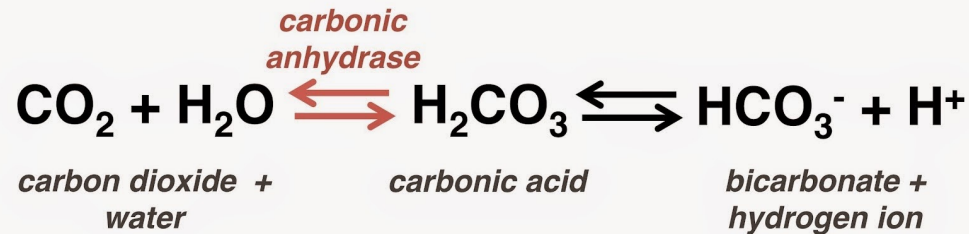


Nearly Instantaneous Balance: Buffering Systems



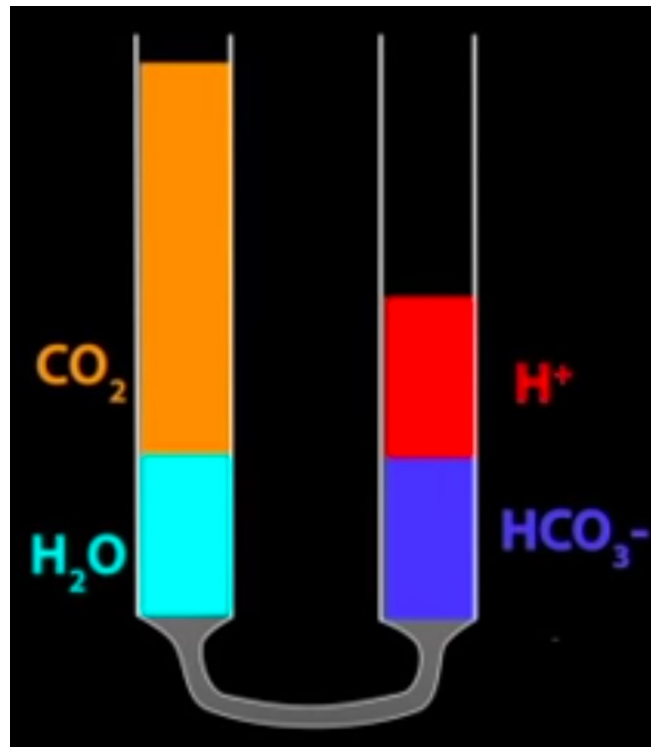
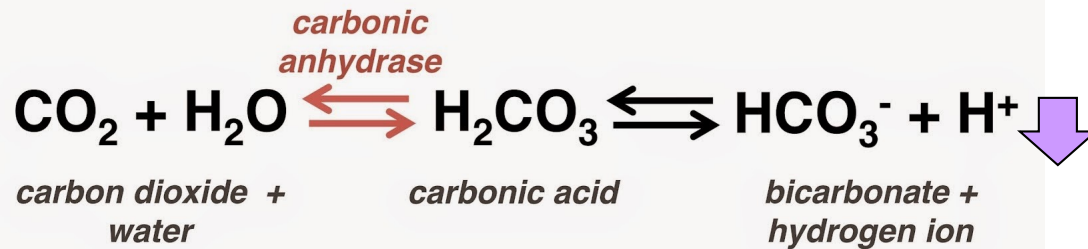
(Cellular Proteins, Hemoglobin, Phosphate Also Act As Buffers)

Nearly Instantaneous Balance: Buffering Systems



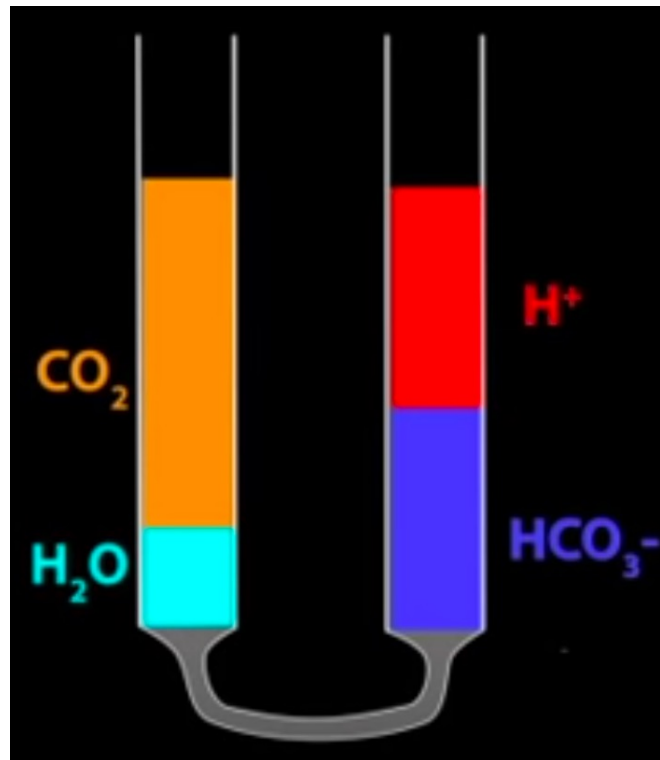
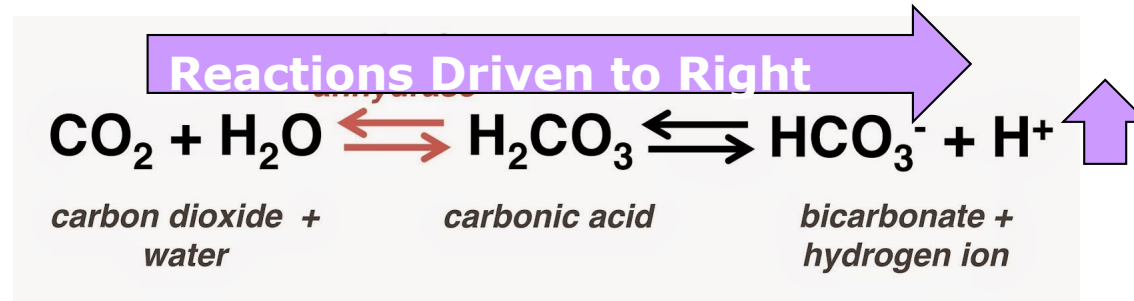
Buffer system will stay balanced. Increases or decreases on one side of equation will cause compensating increase or decrease on other side.

Nearly Instantaneous Balance: Buffering Systems



**Drink alkaline water
and momentarily
decrease H^+**

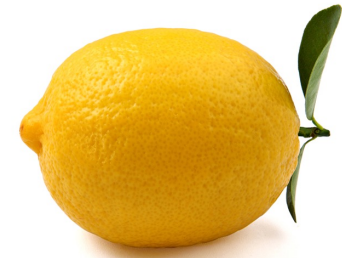
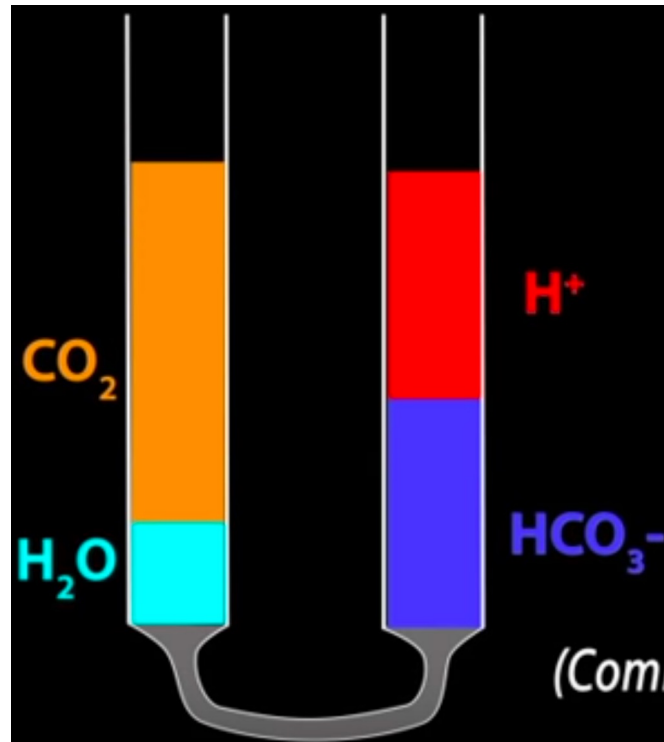
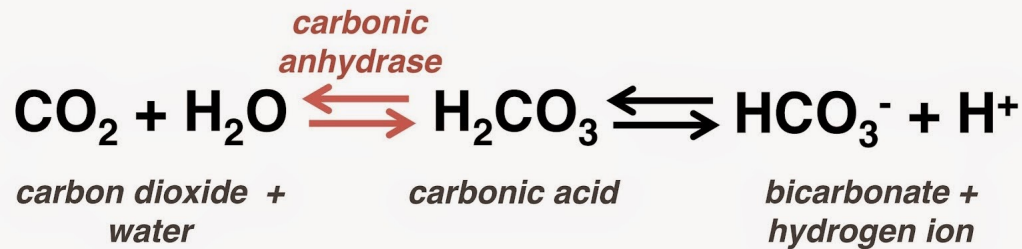
Nearly Instantaneous Balance: Buffering Systems



Some CO_2 automatically converted to carbonic acid and then to H^+ and HCO_3^-

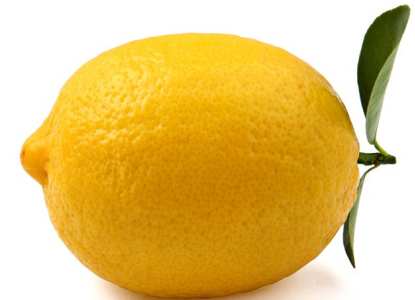
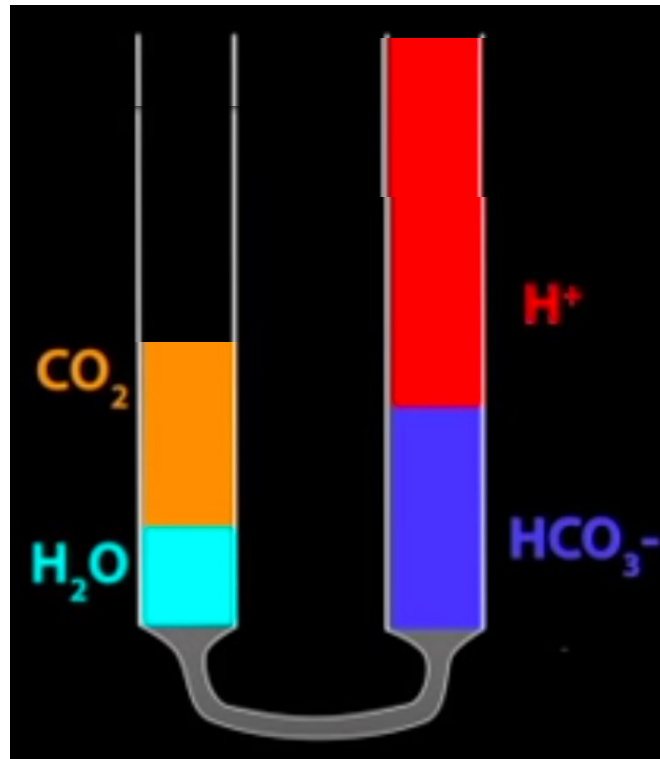
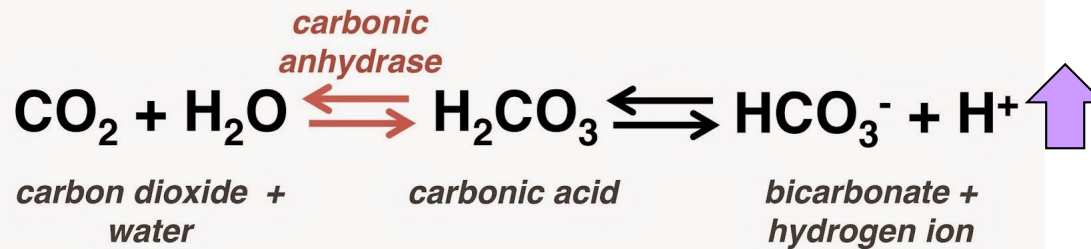
Blood pH instantly returns to more acidic levels where it started!

Nearly Instantaneous Balance: Buffering Systems



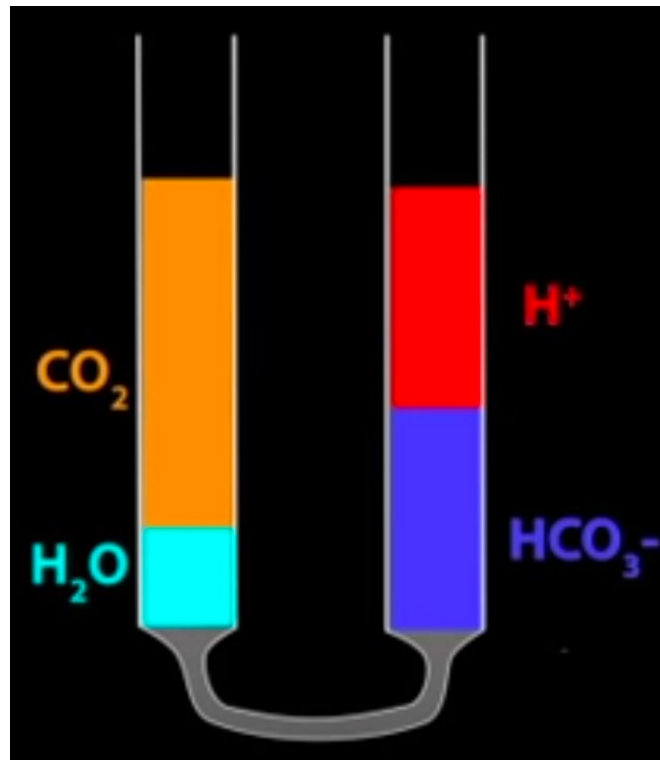
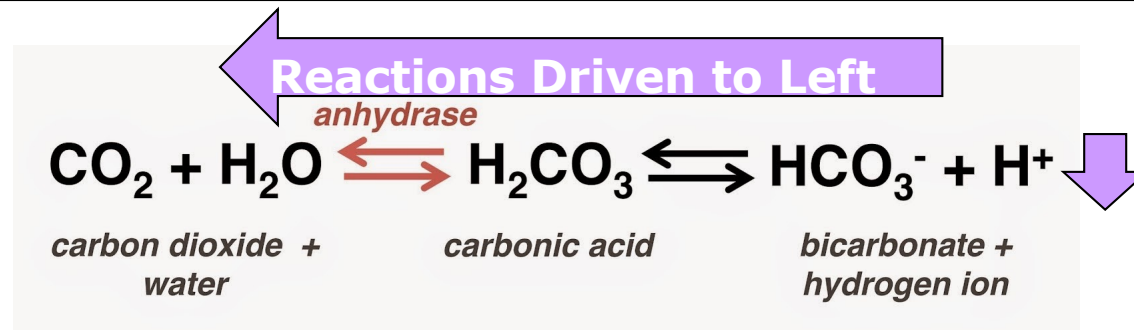
**Eat something very
acidic**
**What would happen
to your body's pH
and how would this
chemical equation
react?**

Nearly Instantaneous Balance: Buffering Systems



**Eat something very
acidic and
momentarily
increase H^+**

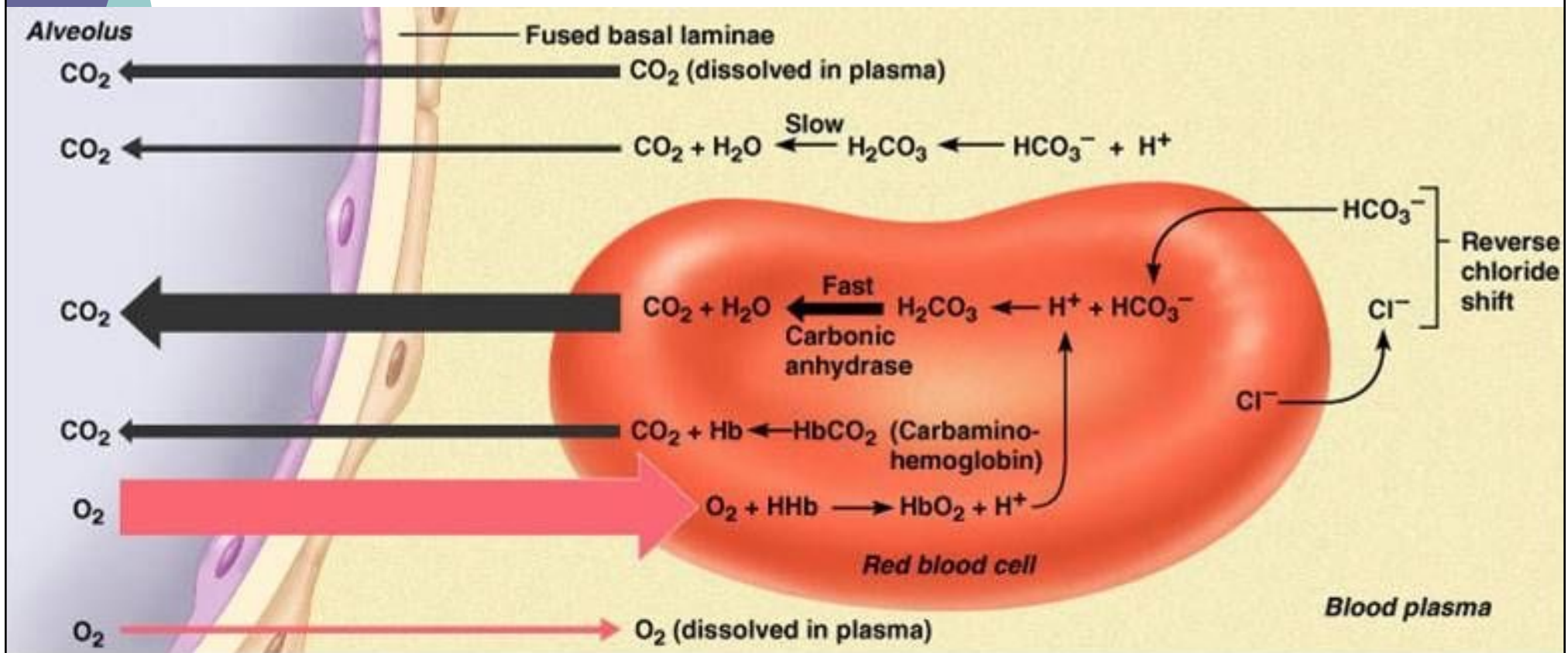
Nearly Instantaneous Balance: Buffering Systems



Some H⁺ combines with HCO₃⁻ to form carbonic acid and then CO₂/H₂O

Blood pH instantly returns to more basic levels where it started!

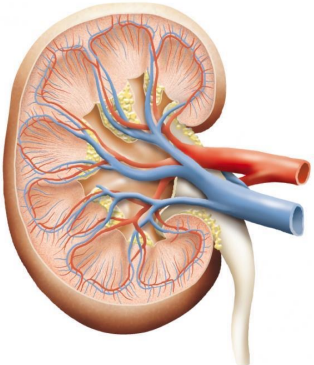
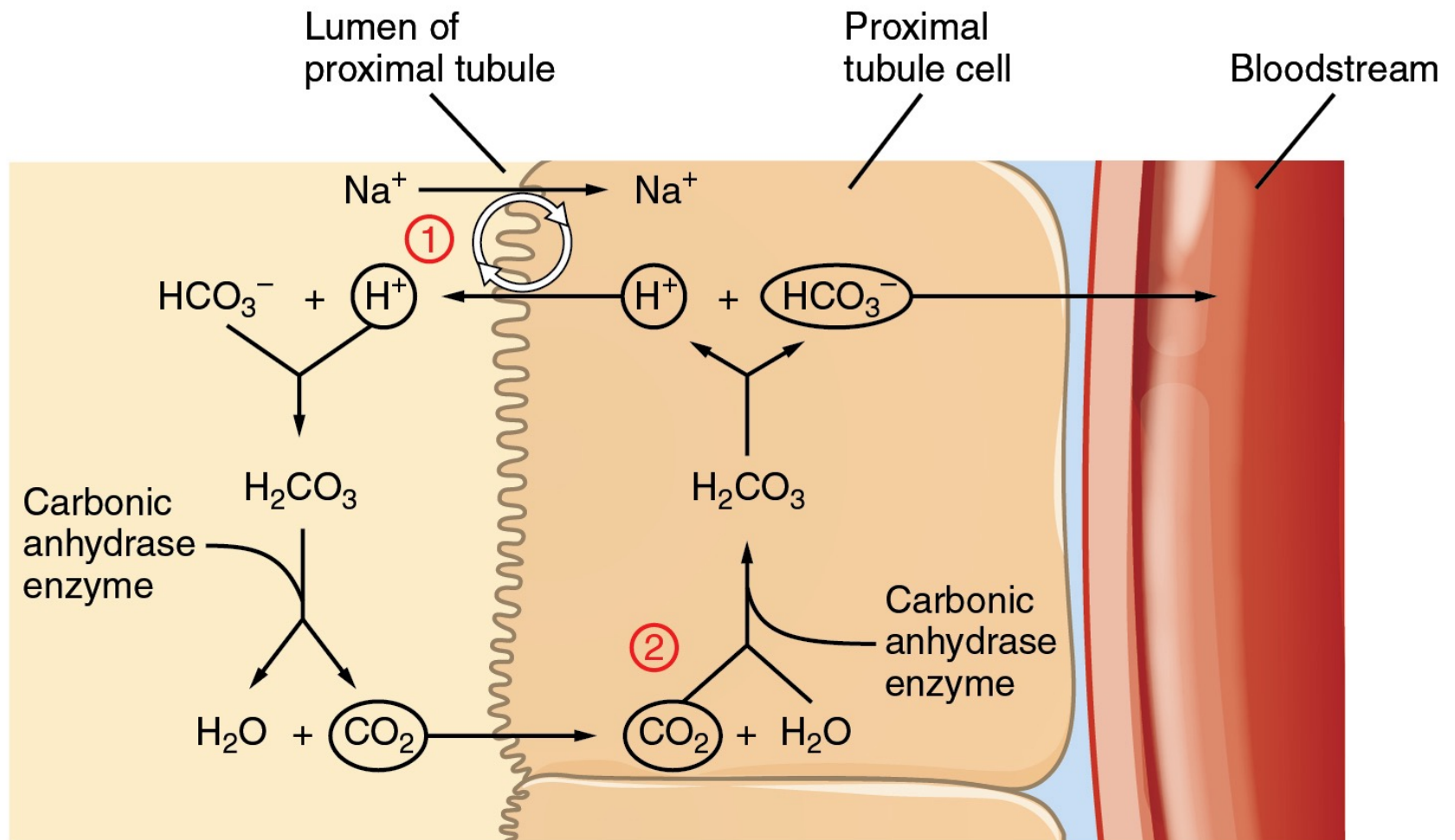
Pretty Quick Balance: Changes in Respiratory Rate



b) Oxygen pickup and carbon dioxide release in the lungs

https://www.youtube.com/watch?v=pl_-P3rM5n8

Longer Term Balance: Kidneys Eliminate or Preserve $\text{H}^+/\text{HCO}_3^-$



A more detailed overview of everything if you like...
https://www.youtube.com/watch?v=hZPeP_oIdXk



The Mysterious Appeal of Alkaline Water

Imagine a friend tells you, “I heard that most people’s bodies are too acidic these days and we need to drink alkaline water to get ourselves back into balance.”

How would you respond? Include in your response:

- **information about our *buffer system***
- **information about the activity of either the *lungs or kidneys*.**